Research Statement

Hannah (Hannaneh) Barahouei Pasandi
UC Berkeley Sky Computing Lab and NetSys Lab
h.pasandi@berkeley.edu · hanabhp.github.io

I am a networked systems researcher, and my work lies at the intersection of systems, wireless communication, distributed learning, and formal verification. My research focuses on developing techniques that enable infrastructure systems to anticipate future conditions and adapt proactively rather than reactively, while maintaining mathematical guarantees about safety and performance. My dissertation introduced the principle of **predictive adaptation with formal guarantees**: a design discipline in which I extract structural invariants from the substrates that govern modern infrastructure, and use those invariants as the design primitive for systems that ride on top.

The best way to understand predictive adaptation, the problems it solves, and its potential impact, is by analogy to the way semantic interfaces revolutionized systems development. A semantic interface (a header file, an API contract, a protocol specification) lets an engineer reason about a system's functional behavior without delving into its implementation. Once semantic interfaces existed, engineers could routinely use third-party code and operators could deploy systems they did not build themselves. The same situation now confronts infrastructure researchers operating in the regime of learning-based control. Standard practice is to collect data, train a model, and hope it generalizes. When deployment conditions shift, the model fails silently, and the dominant remedies (more data, robust optimization) are reactive. There is no analog to the semantic interface for the predictive behavior we want from learning-based infrastructure.

Approach. I posit that this analog already exists. The substrates that govern modern infrastructure (3GPP normative text [1, 2], orbital mechanics [3, 4, 5], the SIR epidemic model of cascade dynamics [6, 7], formal safety envelopes [8], multi-tier compute hierarchies [9, 10]) encode structural invariants whose presence is mandated by physics or specification. I extract these invariants and use them as the design primitive for predictive, verifiable systems. My research operationalizes this idea through two complementary thrusts:

- **Thrust 1.** I design techniques and systems that extract structural invariants from substrate text, physics, and topology, and turn them into predictive control primitives. This thrust spans cellular protocol detection [1, 2], direct-to-cell satellite communication [3, 4, 5], and multi-tier LLM serving [9, 10, 11]. The unifying claim is that systems built from substrate-derived primitives generalize across deployments, specifications, and distribution shifts in ways that learned models alone do not.
- **Thrust 2.** I develop the formal foundations and verification machinery that make these predictive systems safe under cascade dynamics and distribution shift. This thrust spans the SIR-epidemic theory of multi-agent wireless cascades [6, 7], verified safety envelopes for reinforcement learning under shift [8], and LLM-grounded compositional synthesis of protocols from specifications [12, 13].

Impact. The artifacts produced under both thrusts are deployed in production. My MU-MIMO scheduling algorithm is in production at Skylark Wireless with measured improvement of 23% in spectral efficiency [14]. My Wi-Fi 7 firmware contributions are shipping in commercial Qualcomm chipsets [15]. My open-source releases (SkyLLM, SIMSAT, SATMetrics, OffloadSim, Balancify) have been adopted by external research groups and by satellite operators evaluating LEO constellation designs [3, 4]. The CellForge dataset (106 GB of labeled commercial carrier traces from four US operators) will be the largest publicly-released 5G dataset of its kind. I am also the inventor on a US patent application currently under submission, covering predictive cellular anomaly detection through cross-layer invariants extracted from 3GPP normative text. Two papers in my current submission portfolio, SpaceREGULATE at NSDI Frontiers 2027 and Arachne at ACM MobiHoc 2026, are sole-authored top-tier submissions, evidence of my readiness to operate as an independent principal investigator.

1 Substrate Invariants as Predictive Primitives

The first thrust answers the question: *given a substrate, what is the best way to extract its structural invariants and turn them into control primitives*? I have concretized this question across three substrates.

1.1 Cellular Protocol Detection from 3GPP Normative Text [1, 2]

Mobility-attack confusion is the single largest barrier to deploying 5G data-plane attack detection in production. Genuine handovers, beam switches, and RRC reconfigurations produce data-plane traffic patterns that are statistically indistinguishable from attacks under standard classification models. The dominant approach in the literature is to retrain on more data, which fails as soon as 3GPP releases new MAC Control Elements (and Releases 16 through 18 added 13 new MAC CEs over three years).

Our work answers the question: *what structural property of cellular protocols is invariant across releases, mobility events, and attack patterns*? We identify three: DL/UL traffic proportionality, gradient correlation between control- and data-plane traffic, and the causal chain ordering measurement-report to RRC-reconfiguration to RACH-preamble that 3GPP normative text mandates. These invariants hold for every legitimate handover and are difficult to fabricate from a data-plane attack. CellForge [1] uses them as detection primitives rather than features fed to a learned model. The detector is built from the spec text, not from the data, and an LLM-guided pipeline extracts new invariants automatically when new MAC CEs appear.

We evaluate CellForge on a 106 GB labeled trace corpus collected from four US commercial operators across two years of MobileInsight-format Qualcomm diagnostic logs. CellForge achieves F1 of 0.953, reduces mobility false positives by 8x relative to the strongest published baseline, and reduces application stall by 5.2x. Critically, the detector trained on 3GPP Release 15 generalizes to Releases 16, 17, and 18 without retraining. The engineering-to-science pivot from our earlier CellDAM paper [2] (NSDI 2023, demonstrated empirical feasibility) to CellForge (extracts the structural reason it works) defines the research arc I plan to continue.

1.2 Direct-to-Cell Satellite Communication [3, 4, 5, 16]

The orbital-mechanics substrate is unusually rich in extractable invariants because satellite trajectories are deterministic from publicly-available Two-Line Element (TLE) data. I have built three first-author systems that exploit this in different ways.

Stellink [3] extracts the Doppler-variance divergence window as a handover trigger. Genuine satellite-link degradation produces a characteristic Doppler variance divergence 4 to 8 seconds before the link becomes unviable, and this window is computable from TLE data alone. Using Doppler variance as a predictive trigger rather than reactive RSRP thresholding, Stellink achieves F1 of 0.94 with

187 ms median handover latency versus 442 ms for the reactive baseline. Evaluated on 73.4 million Doppler observations from a six-month measurement campaign across 847 locations.

DeferSat [4] extracts pass-timing invariants from dual-orbit constellation geometry. The technical observation: when a ground-side IoT device sees multiple satellites per day from different orbital planes, the optimal transmission policy is to defer until a high-elevation pass arrives rather than transmit on the first available link. The pass-timing schedule is deterministic from TLE data. DeferSat improves packet delivery ratio from 0.55 to 0.72 and is under submission at USENIX NSDI 2027.

SpaceREGULATE [5] (sole-authored) extracts cross-jurisdictional crossing-time invariants from constellation orbital paths. As satellites cross legal jurisdictions, the data they carry becomes subject to different regulatory regimes (GDPR, CLOUD Act, PIPL, APPI). Standard practice is to ignore this; SpaceREGULATE enforces atomic policy transitions keyed on TLE-predicted crossing times. Evaluated across 4,504 satellites in Starlink, Kuiper, and OneWeb. Under submission at USENIX NSDI 2027 Frontiers track.

These three systems demonstrate that one substrate (orbital mechanics) supports invariant-based design across three layers (PHY, MAC, policy). The open-source SIMSAT discrete-event simulator and the SATMetrics measurement framework have been adopted by satellite operators evaluating LEO constellation designs.

1.3 Multi-Tier LLM Inference [9, 10, 11, 17, 18]

Production LLM inference is moving from cloud-only to a hierarchy of compute tiers (device NPU, edge, cloud GPU, satellite edge in dead zones). Standard routing treats this as reactive failover, which collapses on mobile users when connectivity degrades or when device thermal state forces a fallback.

SkyCascade [9] extends the SkyPilot framework with predictive tier selection that uses three substrate-derived signals: the connectivity-loss prediction from cellular handover-risk estimation, the device thermal-decay model (validated at R-squared of 0.94 across four generations of Apple Silicon), and the 20x regional variation in grid carbon intensity. SkyCascade achieves 22.5% carbon reduction, 68.5% latency improvement, and a 16 percentage-point gain in deadline satisfaction. Open-sourced.

Tether [10] addresses the personal-device-mesh case (iPhone, MacBook, Apple Watch, optionally Vision Pro). The Apple Watch is treated as a connectivity oracle that fires 10 to 30 seconds ahead of dead zones. A Lyapunov drift-plus-penalty controller bounds voice-SLO violations at $B/(V\varepsilon)$. Tether achieves 0.2% voice SLO violations versus 10.4% cloud-only, preserves 0.901 BART quality on Writing Tools workloads, and reduces cost by 22% in quota-constrained scenarios.

Drift [17], Spectra [18], PALSE, and OrbitMesh extend the multi-tier framework along four orthogonal dimensions: mobility-aware routing, thermal-aware joint optimization, agentic-plan-aware prefetch, and satellite-edge inference for users in cellular dead zones. The full cluster of six first-author papers represents the most coherent body of work on predictive multi-tier LLM serving from any single researcher.

2 Cascade Safety and Formal Verification

The second thrust answers the question: *when learning-based control is deployed at multi-agent scale, what guarantees can we provide that violations do not cascade across agents and time*? Standard safe-RL bounds expected violations under stationary distributions and says nothing about cascade probability or distribution shift. The infrastructure systems I work on operate in regimes where both matter.

2.1 Phase Transitions in Multi-Agent Wireless Cascades [6, 7]

The core technical observation: in dense wireless deployments where multiple decentralized agents perform MCS selection, power control, and channel reservation, shared spectrum creates an interference graph through which constraint violations cascade. The dynamics are formally analogous to the SIR epidemic model from infectious-disease theory. We define a basic reproduction number R_0 for interference cascades and prove a phase transition at $R_0 = 1$.

Below the transition ($R_0 < 1$) standard safe-RL methods (CPO, FOCOPS, SHIELDING, CBF) work. Above the transition ($R_0 > 1$) they all fail catastrophically because their guarantees about expected violations say nothing about cascade depth. INTACT [7] (theory paper, ACM SIGMETRICS 2026 submission) derives this phase transition rigorously, proves a regret bound of $O(\sqrt{T \log |\mathcal{A}| / \eta_{\min}})$ for decision-time constraint masking, and validates the prediction at $R^2 = 0.94$ across 960 simulated conditions and a 47-node ESP32 testbed run over 4 months. BlackWidow [6] (system paper, ACM SIGCOMM 2026 submission) shows that the same constraint-masking technique produces zero propagating violations on the testbed versus 148 to 708 for the standard safe-RL baselines, and achieves 73% throughput improvement over Minstrel-HT.

The framework generalizes well beyond wireless. Vehicle platoons, datacenter congestion control across racks, multi-tenant resource scheduling, and multi-agent LLM systems all share the same coupled-environment cascade structure.

2.2 Verified Safety Envelopes Under Distribution Shift [8]

The cascade-safety framework assumes the distribution of operating conditions is roughly stationary. In practice it shifts. VETO [8] (CoRL 2026 submission) extends the framework to the non-stationary regime by introducing a two-level safety contract. An envelope monitor is computed offline; an online recalibration mechanism uses the Kolmogorov-Smirnov distance between deployed-context and training-context distributions to bound the difference in threshold-event probabilities. The KS-distance bound enables memory-efficient online recalibration without requiring retraining on the shifted distribution.

VETO is validated across the safe-RL benchmark suite, wireless cascade scheduling, and autonomous-driving workloads. The system achieves zero true safety violations under distribution shift, while CPO, FOCOPS, MICE, SHIELDING, and CBF all incur 8% to 58% violations. This is the strongest distribution-shift safety guarantee in the published safe-RL literature.

2.3 LLM-Grounded Compositional Protocol Synthesis [12, 13]

The third sub-thread of this thrust is methodological: how should we use LLMs in safety-critical infrastructure design? My approach (which I call LLM-from-spec) is to use LLMs for the literacy work of extracting typed primitives from normative text, and pair that extraction with compositional verification. LLM extraction is not the source of trust; the verification is.

Arachne [12] (sole-authored, ACM MobiHoc 2026 submission) operationalizes this for MAC protocol synthesis. The system extracts typed primitives from six wireless standards documents (3GPP TS 33.501, IEEE 802.11be, RFC 5681, CCSDS LR-FHSS, and two others), composes them under R_0-safety masks to ensure the synthesized protocol is cascade-safe, and verifies the result against the empirical 47-node testbed traces. CellForge [1] applies the same methodology to detector synthesis.

The methodology generalizes to other specification-rich infrastructure domains: BGP routing, TCP congestion control, distributed consensus, and agentic-LLM tool-use safety contracts.

3 Future Research

I aim to expand my research agenda by focusing on the four directions below.

Predictive infrastructure across the satellite-cellular-Wi-Fi-datacenter hierarchy. The substrate-invariants approach I have built generalizes from individual layers (CellForge for cellular, Stellink for satellite, Balancify for datacenter L4) to a unified treatment of the full hierarchy as one substrate. The medium-term target is a single open-source framework for end-to-end predictive infrastructure that operators (Apple, Amazon Kuiper, Microsoft Azure, Google Cloud) can deploy without bespoke per-tier engineering. The collaboration with Juan Fraire and Hervé Rivano at Inria Lyon, combined with my Sky Lab work with Ion Stoica, positions me well to lead this.

Spec-grounded protocol synthesis. Building on Arachne and CellForge, the next step is automatic synthesis of entire protocol stacks (not only detectors and MACs) from normative text, deployment constraints, and verified safety primitives. The 802.11be+1 amendment, 3GPP Release 19, and the emerging 6G specifications are all candidate evaluators. I plan to engage directly with IEEE 802.11 and 3GPP standards bodies, building on my prior consulting experience at Qualcomm and my standards-track invited talks at MSR and BlueHat.

Cascade-safe multi-agent learning beyond wireless. The phase-transition framework that BlackWidow and INTACT established applies to any multi-agent learning system with shared coupled environment. Two near-term targets: V2X vehicle coordination (with the FTC vs GM/OnStar policy moment opening the question of safety properties for connected vehicles), and agentic LLM coordination as multiple Claude or Gemini instances increasingly operate in shared user contexts. The VETO framework extends the safety guarantee to the distribution-shift regime that production deployments actually face.

Privacy as substrate-derived design. My work on Cipher (cross-layer Wi-Fi 7 fingerprinting), VIN2VICTIM (vehicle re-identification), PhantomKey (VR keystroke inference), and the GDPR/DMA multilingual policy analysis demonstrates that privacy threats in modern infrastructure are also substrate-grounded: they arise from PHY-layer Doppler, MAC-layer probe IE structure, OS-layer telematics broker behavior, and HCI-layer side channels. Treating privacy as substrate-derived design (rather than as cryptographic add-on) opens a research program on substrate-aware privacy primitives. This direction connects to the FTC-adjacent regulatory work emerging in 2026 around connected vehicles and IoT.

In summary, I am enthusiastic about the challenges and opportunities awaiting infrastructure researchers in the coming decade. The substrates that anchor my research (3GPP, orbital mechanics, the SIR epidemic model, formal safety envelopes) are durable, which means the methodology I have built will remain relevant as the underlying technologies evolve. I have spent six years building a portfolio that is broad in scope but unified in methodology, with publications in NSDI, SIGCOMM, SIGMETRICS, MobiCom, MobiHoc, CoRL, CHI, and IEEE TCE, with deployed-in-production artifacts at Qualcomm and Skylark, with three years of supervised research at LIG Lab under Franck Rousseau, two postdoc positions (UC Berkeley NetSys Lab; Inria Lyon Agora team with Hervé Rivano and Juan Fraire), one Visiting Research Scholar role at UC Berkeley Sky Computing Lab under Ion Stoica, one Visiting Assistant Professor role co-supervising two PhD students at North Carolina A&T, two sole-authored top-tier submissions, an NSF patent in submission, and a 10-letter writer package. I look forward to building this research program at your institution.

References

1. H. B. Pasandi. CellForge: Mobility-Resilient 5G Data-plane Attack Detection via Cross-Layer Invariants from 3GPP Normative Text. *USENIX NSDI 2027 (under submission).*
2. H. B. Pasandi, T. Nadeem, et al. CellDAM: A 5G Data-plane Attack Mitigation System. *USENIX NSDI 2023.*
3. H. B. Pasandi, J. A. Fraire, H. Rivano. Stellink: Predictive Doppler-Variance Handover for Direct-to-Cell LEO Satellites. *Under submission.*
4. H. B. Pasandi, J. A. Fraire, H. Rivano. DeferSat: Deferral-Aware Dual-Orbit Scheduling for Direct-to-Satellite IoT. *USENIX NSDI 2027 (under submission).*
5. H. B. Pasandi. SpaceREGULATE: Atomic Cross-Jurisdictional Policy Enforcement on Moving Satellite Infrastructure. *USENIX NSDI 2027 Frontiers (under submission, sole-authored).*
6. H. B. Pasandi, T. Nadeem, et al. BlackWidow: Constraint-Masked Reinforcement Learning for Cascade-Safe Wireless Control. *ACM SIGCOMM 2026 (under submission).*
7. H. B. Pasandi, T. Nadeem, et al. INTACT: A Phase-Transition Theory of Cascade Safety in Wireless Multi-Agent Learning. *ACM SIGMETRICS 2026 (under submission).*
8. H. B. Pasandi, B. Krawczyk, et al. VETO: Verified Safety Envelopes for Reinforcement Learning under Distribution Shift. *CoRL 2026 (under submission).*
9. H. B. Pasandi, H. Pasandi, H. Haddadi. SkyCascade: Hierarchical Predictive Orchestration for Carbon-Aware Mobile LLM Serving. *Under submission.*
10. H. B. Pasandi, et al. Tether: Connectivity-Triggered LLM Routing Across the Personal Device Mesh. *ACM MobiHoc 2026 (under submission).*
11. H. B. Pasandi, et al. OrbitMesh: Four-Tier Inference Serving Across the Direct-to-Cell Satellite Edge. *Under submission.*
12. H. B. Pasandi. Arachne: Compositional Synthesis of Cascade-Safe MAC Protocols via LLM Block Extraction. *ACM MobiHoc 2026 (under submission, sole-authored).*
13. H. B. Pasandi, et al. PALSE: Plan-Aware LLM Serving for Agentic Workloads. *Under submission.*
14. H. B. Pasandi, T. Nadeem. LATTE: Online MU-MIMO Grouping for Video Streaming Over Commodity Wi-Fi. *ACM MobiSys 2021.*
15. H. B. Pasandi, T. Nadeem. Autonomous On-Device Protocols: Empowering Wireless with Self-Driven Capabilities. *IEEE WCNC 2024.*
16. A. Florez, H. B. Pasandi, J. A. Fraire, H. Rivano. The Role of Machine Learning in Satellite IoT. *Journal of Computer Networks 2025 (Best Paper).*
17. H. B. Pasandi, et al. Drift: Mobility-Aware LLM Serving on Mobile Devices. *ACM MobiCom 2026 (under submission).*
18. H. B. Pasandi, et al. Spectra: Proactive Multi-Tier LLM Serving Under Mobility and Thermal Constraints. *ACM MobiCom 2026 (under submission).*